

BERKAY BENLI

[LinkedIn](#) — [GitHub](#) — [Personal Website](#)

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EDUCATION

Bachelor of Science (Honours) in Computing Science, University of Alberta (expected Winter 2027)

Relevant Courses

- **Object Oriented Programming, Data Structures and Algorithms, Programming in C**, Calculus I, Calculus II, Linear Algebra

EXPERIENCE

UASTarr Alberta Rocket Research

Jan 2024 - June 2024

Software Team Member

Edmonton, AB

- Collaborated with team members to **design and develop the UASTARR website**, addressing bugs, implementing new features, and **enhancing overall functionality** to improve user experience.
- Contributed to the **development of the rocket tracker software**, **enhancing tracking accuracy by 12%** through the implementation of more **efficient tracking algorithms** such as **Kalman Filter Algorithm**, and **optimizing code efficiency**

PROJECTS

Sample-Searcher [App Link](#) — [Backend Code](#) — [Frontend Code](#) (React, NodeJS, Express, MySQL, Python, AWS)

- Developed a **full-stack** music application using React and NodeJS/Express with a MySQL database, integrating **Python web-scraped data** into a **MySQL database of over 2000 songs**.
- Deployed the backend on **AWS EC2** with **Docker containers** and implemented **CI/CD pipelines**, ensuring seamless updates and scalability.
- Created **RESTful API endpoints** for song retrieval and management. Designed a **responsive UI** with **dynamic song display** and pagination.

Beat-Sequencer — [App Link](#) — [Source Code](#)

(Python, Pygame)

- Designed a Beat Sequencer using **Python** and **Pygame** with a **graphical user interface** with interactive buttons and sliders for beat creation, tempo adjustment, and dynamic grid of pads for composing beats.
- Implemented **sound mapping** and playback, allowing users to toggle pad states, create, play, and clear beats in real-time, with **adjustable difficulty modes**.
- Ensured a responsive user experience through precise mouse click detection and event handling.

EXTRA-CURRICULAR ACTIVITIES

- Competed in UAPC (**University of Alberta coding competition**) multiple times, the Beaver Computing Challenge (Mathematics Contest), and the **University of Waterloo Gauss Contest**, earning an **Honours with Distinction award**.
- Awarded **3rd place** in MecSimCalc **hackathon LIVE** 2023

SKILLS

Computer Skills and Languages: Python, C, MySQL, JavaScript, React, NodeJS, HTML, CSS, Latex, Microsoft Office, Google Suite. Fluent in English and Turkish